

ABRAHAM SCARIYA

Kochi, India

+9074362768 | abraham.s23@iiits.in | [linkedin.com/in/abraham-scariya](https://www.linkedin.com/in/abraham-scariya)

PROFILE SUMMARY

B.Tech Hons Electronics and Communication Engineering student at IIIT Sri City with a strong foundation in VLSI design, embedded systems, and IoT. Certified VLSI Engineer with hands-on experience in CMOS fabrication, RTL design, and full-chip design flow. Proficient in SystemVerilog, Verilog, and C++, with a track record of developing innovative hardware-software integrated solutions and mobile applications.

EDUCATION

Indian Institute of Information Technology Sri City

B.Tech with Honours in Electronics and Communication Engineering

CGPA: 8.33/10

Aug 2023 – 2027

Sri City, Andhra Pradesh

EXPERIENCE

Summer Research Intern – Cognitive Radar Prototype Development

May 2026 – Jul 2026

National Institute of Technology (NIT) Calicut

Calicut, India

- Developing a prototype for **Cognitive Radar** systems utilizing FPGA platforms.
- Implementing **Deep Learning models** onto FPGAs to enable advanced radar signal processing.
- Designing and synthesizing custom radar waveforms using **Verilog**.

VLSI Engineering Trainee

May 2025 – Aug 2025

SMEC Labs

Kochi, India

- Gained hands-on experience in **CMOS fabrication** and VLSI physical design.
- Developed proficiency in SystemVerilog, Verilog, and VHDL for design and verification.
- Executed full chip design flow (RTL to GDSII) and performed simulations using T-Spice, L-Spice, and LTSpice.

PROJECTS

Dynamic Partial Reconfiguration of YOLO Models in FPGA – Honours Project

Aug 2025 – Present

- Implementing **Dynamic Partial Reconfiguration (DPR)** of YOLO models on FPGA drones to minimize power consumption.
- Developing a **hierarchical model** to identify and eliminate data redundancy prior to processing.
- Optimizing the inference pipeline for real-time edge computing in aerial surveillance applications.

Fully Autonomous Drone & Base Station System – ISRO IROC Challenge

Dec 2025 – Present

- Designing a **fully autonomous drone** capable of operating in a **Mars-like simulated environment** for extreme terrain navigation.
- Integrating **LiDAR, depth sensors, IMU, and camera modules** for real-time localization, obstacle avoidance, and telemetry transmission.
- Developing an intelligent **base station** enabling autonomous drone docking, **wireless charging**, and parallel high-speed data transfer.
- Implementing robust navigation algorithms and system-level hardware-software co-design for reliable operation in harsh environments.

FPGA-Accelerated Smart Agriculture CNN – AMD & BITS Pilani Hackathon

Mar 2026

- Designed a fully accelerated custom CNN processing 7 environmental inputs (temperature, humidity, water level, pH, N, P, and K) to predict optimal crop selection.
- Integrated automated actuation logic to dynamically control a water pump based on real-time model inference for efficient irrigation.
- Achieved high-throughput execution by utilizing PS-PL hardware-software co-design on a PYNQ board for edge AI applications.

Hardware-Accelerated MNIST Digit Recognition – PYNQ Board

Feb 2026

- Developed a hardware-accelerated Machine Learning model trained on the MNIST dataset to accurately detect and classify handwritten digits.
- Engineered a custom processing pipeline utilizing PS-PL integration to automatically detect and mathematically add recognized numbers in real-time.

Optimized system performance and inference latency for edge deployment.	
Fabrication of MoS2 Photodetector – <i>Semiconductor Fabrication</i>	Mar 2026 – Present
- Fabricating a highly sensitive MoS2 photodetector device on a Si/SiO₂ substrate. - Performing Molybdenum (Mo) deposition via DC Sputtering, followed by oxidation to MoO₃ and subsequent sulphuration at 550°C in a CVD furnace. - Patterning Aluminum Interdigitated Electrodes (IDE) using thermal evaporation and conducting rigorous device testing.	
6T SRAM RTL to GDSII Flow – <i>Cadence Virtuoso</i>	Feb 2026 – Apr 2026
- Executing the complete RTL to GDSII flow for a 6T SRAM cell architecture. - Performing schematic design, layout generation, and physical verification (DRC, LVS). - Optimizing cell design for area, power, and delay metrics using industry-standard Cadence tools.	
CMOS Inverter & Counter Design – <i>Cadence Virtuoso, RTL to GDSII</i>	Jan 2026 – Feb 2026
- Designed and simulated a CMOS inverter and synchronous counter at transistor and RTL level using Cadence tools. - Completed the full RTL to GDSII flow including schematic design, layout, DRC, LVS, and post-layout simulations. - Analyzed power, delay, and area trade-offs using Cadence tools.	
FusionPass – <i>Raspberry Pi, OpenCV, RFID</i>	Sep 2024 – Dec 2024
- Developed a biometric attendance system combining RFID and facial recognition for enhanced security. - Implemented real-time processing on Raspberry Pi with an Android app for status displays and Excel export.	
AquaVitals – <i>Arduino Uno, NodeMCU, Sensors</i>	Nov 2024 – Dec 2024
- Engineered a smart aquarium system featuring PID-controlled pH regulation for optimal water quality. - Integrated customizable feeding schedules and remote control capabilities via a dedicated app.	
SmartLumin – <i>Arduino, NodeMCU</i>	Aug 2024 – Oct 2024
- Designed a remote-controlled streetlamp system with automatic brightness adjustment based on ambient light. - Implemented a Master-Slave setup utilizing I2C communication for synchronized control.	
TECHNICAL SKILLS	
Programming Languages: Verilog, SystemVerilog, VHDL, C, C++, Embedded C, Python, Kotlin Tools & Software: Cadence, Xilinx Vivado, Intel ModelSim, LTSpice, Arduino IDE Hardware: Arduino, NodeMCU, Raspberry Pi, RFID Modules, Artix, Zynq, Kria, Pynq boards	
TRAININGS & CERTIFICATIONS	
App Development with Kotlin <i>Udemy (Online)</i>	May 2024 – Jul 2024
- Built Android applications using Kotlin and Google Firebase. - Focused on clean architecture, scalability, efficiency and security.	
HONORS & AWARDS	
Winner, ElectroForge Hackathon 2024 — IIIT Sri City	2024
Secured 1st place in institute-level hackathon focused on embedded systems & IoT.	
ISRO IROC Challenge — Selected for Round 2 (Ongoing)	2025
- Selected into the second round of ISRO's IROC challenge for autonomous aerial systems. - Received 6 Lakhs in funding to develop a drone and base station prototype for operations in martial terrain environments. - Working on system-level design, hardware integration, and real-time communication between aerial and ground units.	
LEADERSHIP & ACTIVITIES	
Lead, E-CELL — IIIT Sri City	Jul 2025 – Present
Director of Operations, E-CELL — IIIT Sri City	Jul 2024 – Jun 2025
Organizing Committee Member, Ascenta 2025 — IIIT Sri City	2025